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(An Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE New Delhi)

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Department of Computer Science and Engineering



Societal Project Synopsis

III Semester - Societal Project (1BSOP308) — 2025 Scheme

Academic Year 2026-27

Batch No.: 2	Section: F	Guide Name: Prof. Indrakumar D R, Assistant Professor	Submission Date: 29/09/2026
Societal Project Title: EduSolve			

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Project Category		Education, Digital Literacy and Skill Development

Signature of Guide

Abstract

EduSolve is a technology-based platform designed to connect students with real-world problems identified by communities, non-governmental organizations, local organizations, and other problem owners. Students often gain technical knowledge through coursework and projects, but may not have a structured mechanism to discover meaningful societal problems that match their skills and interests. At the same time, community stakeholders may have practical problems that could benefit from student participation, technical knowledge, and collaborative problem-solving. EduSolve proposes a web-based platform where problem owners can post challenges and required skills, students can discover suitable opportunities, and the system can support skill-based recommendations, student interest, owner approval, communication, and creation of a collaborative project workspace. The proposed prototype will be validated with selected stakeholders using measures such as usability, recommendation relevance, and successful problem-to-student/team connections. The expected outcome is a structured bridge between academic learning and practical societal problem-solving, enabling students to gain meaningful project experience while supporting communities in identifying potential student-led solutions.

Introduction

Education increasingly emphasizes practical skills, critical thinking, experiential learning, and problem-solving. Project-based and community-engaged learning approaches provide opportunities for students to apply classroom knowledge to authentic situations and develop technical as well as transferable skills [1]-[4]. However, connecting students to externally identified community problems is not always a structured part of the learning process. Research on engineering community engagement also identifies practical challenges such as logistics, communication gaps, and limited attention to community-partner perspectives [2], [12].

EduSolve addresses this need by creating a single digital pathway from *real-world problem* to *student skill matching* to *team collaboration*. The platform is intended as a prototype for the selected stakeholder group rather than a full-scale deployment of community solutions.

Motivation

Field Observation

Initial team discussions and project-context observations indicate two recurring needs: students want meaningful opportunities to apply technical skills, while problem owners may not have a structured way to reach students with appropriate skills. These observations will be validated through the planned stakeholder survey and field study.

Data / Reports

University Grants Commission guidance encourages higher-education institutions to connect learning with community engagement, practical exposure, and collaboration. India's National Education Policy (NEP) 2020 also emphasizes experiential, inquiry-based, and problem-solving-oriented learning. These policy directions support the relevance of a structured mechanism that connects academic skills with real community needs [21], [22].

Why Us?

The project team is composed of Computer Science and Engineering students with experience and coursework in software development, web technologies, databases, and emerging artificial-intelligence technologies. EduSolve provides a practical context in which these skills can be applied to a socially relevant problem through stakeholder interaction, prototype development, and validation.

Problem Statement

Students may possess academic and technical skills but lack a structured mechanism to discover and collaborate on meaningful real-world problems, while communities and organizations may have practical problems that require student participation and technical support. Therefore, a digital platform is needed to connect real-world problems with suitable students based on skills and interests and to support the subsequent approval and collaboration process.

Stakeholders

- Students
- Communities and non-governmental organizations
- Local organizations and problem owners
- Educational institutions

Scope

The prototype includes problem posting, skill-based recommendations, student interest/expression, owner approval, communication, and project-workspace creation. It excludes financial transactions, full-scale implementation of community solutions, and deployment beyond the selected validation group.

Objectives

1. To survey **50 students and 10 problem owners in Bengaluru** to assess the need for structured student-community project collaboration.
2. To develop an **EduSolve prototype** supporting problem posting, skill-based matching, student interest, owner approval, communication, and collaborative project workspaces.
3. To validate the prototype with **20 students and 5 problem owners** and target at least **80% positive usability feedback**.

Sustainable Development Goal

SDG 4 - Quality Education

Target 4.4 - Relevant skills for employment, decent work and entrepreneurship [23].

EduSolve supports this target by giving students opportunities to apply academic and technical skills to practical, real-world problems through collaborative project work. The platform is therefore aligned with skill development and experiential learning.

Expected Outcome

- **Prototype:** A working EduSolve web prototype for problem posting, skill matching, approval, communication, and project collaboration.
- **Societal impact:** A structured pathway for students to gain practical experience while communities can connect with potential student contributors.
- **Publication:** A research paper targeting a Scopus/Web of Science indexed conference or journal.
- **IPR and report:** Final project report and copyright/IPR submission, if applicable.

Note: Survey numbers above are planned targets and will be replaced by actual collected results where required.

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